Analog-to-Digital Conversion (ADC)

CSE 3105 - Computer Interfacing & Embedded System

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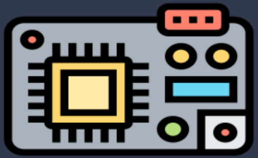


Analog-to-Digital Converter

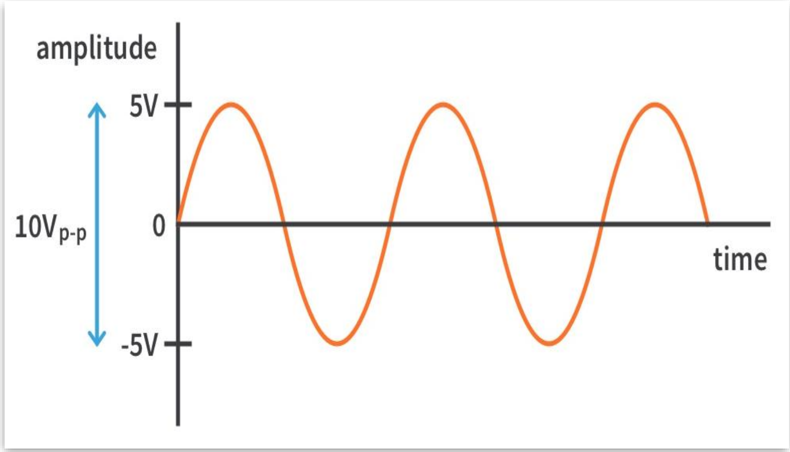
- An **Analog to Digital Converter (ADC)** is an electronic component or system that converts continuous analog signals (like temperature, sound, or light intensity) into a discrete digital representation that can be processed by digital systems, such as microcontrollers or computers.
- It produces a **finite-precision signed or unsigned digital number** to represent approximately the size of an analog voltage relative to a reference voltage.

Analog-to-Digital Converter

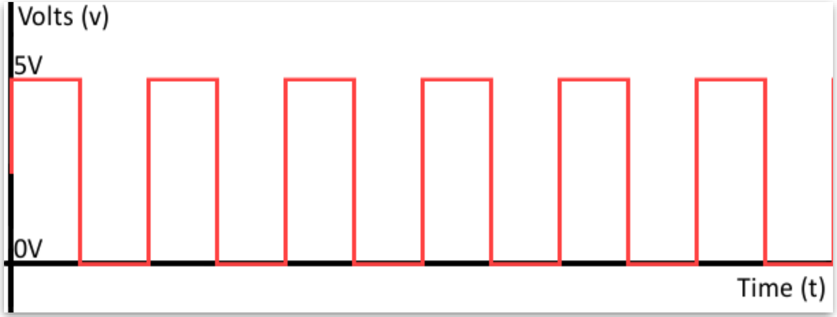
- The reference voltage is a fixed voltage provided by the internal circuit of the microprocessor or by an external circuit connected to a pin of the microprocessor.
- **It does not convert a voltage larger than the reference voltage.**

Analog Signal and Digital Representation

An analog signal is a continuous signal that represents physical phenomena (e.g., voltage, temperature, sound). It can take any value within a given range.



A digital signal is discrete and represents analog data in binary form (0s and 1s), which can be processed by computers or digital systems..



Working Procedure of ADC

- **Sampling**: The analog signal is sampled at regular intervals. The frequency at which this is done is called the sampling rate.
    - According to the Nyquist Theorem, the sampling rate must be at least **twice** the highest frequency of the analog signal to accurately capture it.
    - It indicates how many conversions an ADC performs in a second. *An ADC can perform up to several million or billion samples per second.*

Working Procedure of ADC

- **Quantization**: Each sampled value is approximated to the nearest discrete level. The number of discrete levels is determined by the resolution of the ADC. **The resolution of the ADC is the number of bits used to represent the analog value.** The more bits, the higher the resolution, and the more precise the conversion.
  - While standard resolutions vary between 6 to 24 bits, the resolution has not been improved much in the past few years because a 12-bit or 24-bit resolution is often sufficient for most modem applications.

Working Procedure of ADC

- **Quantization Error:** Due to the rounding of the analog signal to the nearest discrete value, there is always some error introduced. This is called quantization error.
- **Reference Voltage (V_ref)**: The ADC converts the input signal to a digital value relative to a reference voltage (V_ref). This reference voltage defines the range of input voltages the ADC can convert.
  - *For example, if the V_ref is 3.3V and the ADC is 10-bit, the ADC converts input voltages between 0V and 3.3V into a digital value between 0 and 1023.*
- The power dissipation measures the power efficiency of ADC converters.

ADC Architectures

- There are three most popular ADC architectures: sigma-delta ADC for low-speed applications, successive-approximation (SAR) ADC for low-power applications, and pipelined ADC for high-speed applications.
- Sigma-delta ADCs are mostly used in applications requiring low sampling rates, but high resolution, typically less than 100 kilo samples per second and 12 to 24 bit resolution, such as voice band and audio applications. They have been widely used in modern cell phones.

ADC Architectures

- SAR ADCs are suitable for applications with low-power data acquisitions and moderate sampling rates, typically less than 5 million samples per second (MSPS).
- Pipelined ADCs are widely used for high-speed applications, such as digital oscilloscopes, HDTV, and radar communication, requiring fast sampling rates greater than 5 MSPS and relatively low resolution less than 18 bits.

ADC Architectures

The ADC on STM32 microcontrollers is based on the successive-approximation (SAR) architecture. The architecture includes two major components: *sample-and-hold amplifier (SHA)*, and *SAR digital quantization.*

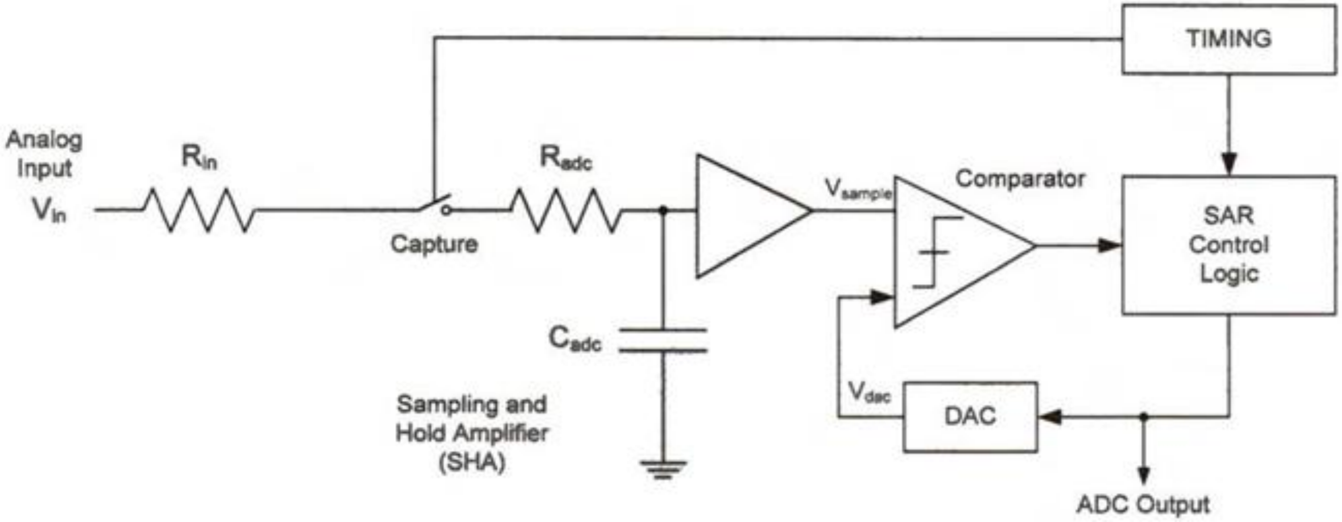


Fig: Basic architecture of successive-approximation (SAR) ADC

SAR Digital Quantization

The SAR control logic uses the binary search algorithm to find the digital number (ADC output) that represents the analog input most closely. The SAR control logic dynamically changes the ADC output so that the digital to-analog converter (DAC) output V_{dac} gradually approaches the DAC input voltage V_{in}.

(1) The conversion starts with setting the internal DAC output V_{dac} to $\frac{1}{2}V_{REF}$ and then compares the DAC output V_{dac} with the ADC input V_{in}.

(2) If V_{in} is larger than $\frac{1}{2}V_{REF}$, the SAR logic controller sets the most significant bit (MSB) of the ADC result. Otherwise, the MSB of the ADC is cleared.

(3) Next, the DAC output V_{dac} is set to either $\frac{3}{4}V_{REF}$ or $\frac{1}{4}V_{REF}$, depending on the comparison result between V_{in} and V_{dac}.

(4) This process repeats until all bits of the ADC output have been determined.

ADC Conversion Process

- Suppose you have an 8-bit ADC with a reference voltage of 5V, and you are trying to convert a 3V input signal.

The resolution is $\frac{\text{V\_ref}}{2^N} = \frac{5V}{256} \approx 0.01953\,V$ per step.

The digital value corresponding to a 3V input is:

$$\text{Digital Value} = \frac{3V}{0.01953V} \approx 153 \text{ (out of 255)}$$

ADC Conversion Process

Lets convert an input voltage of 0.690V into a 4-bit binary value by using SAR ADC. Suppose the range of the input voltage is between OV and lV.

(1) After the input voltage is sampled, the SAR control logic starts the conversion by setting V_{dac} as 0.5V (*i.e.*, the ADC output is `0b1000`) and comparing it with V_{sample}.

(2) Since V_{sample} is larger than V_{dac}, the control logic then sets V_{dac} as 0.750V (*i.e.*, the ADC output is `0b1100`).

(3) Because V_{sample} is smaller than V_{dac}, the control logic then sets V_{dac} as 0.625V.

(4) The above process repeats and the final output of the ADC is `0b1011`. In this example, the conversion process takes four cycles.

ADC Conversion Process

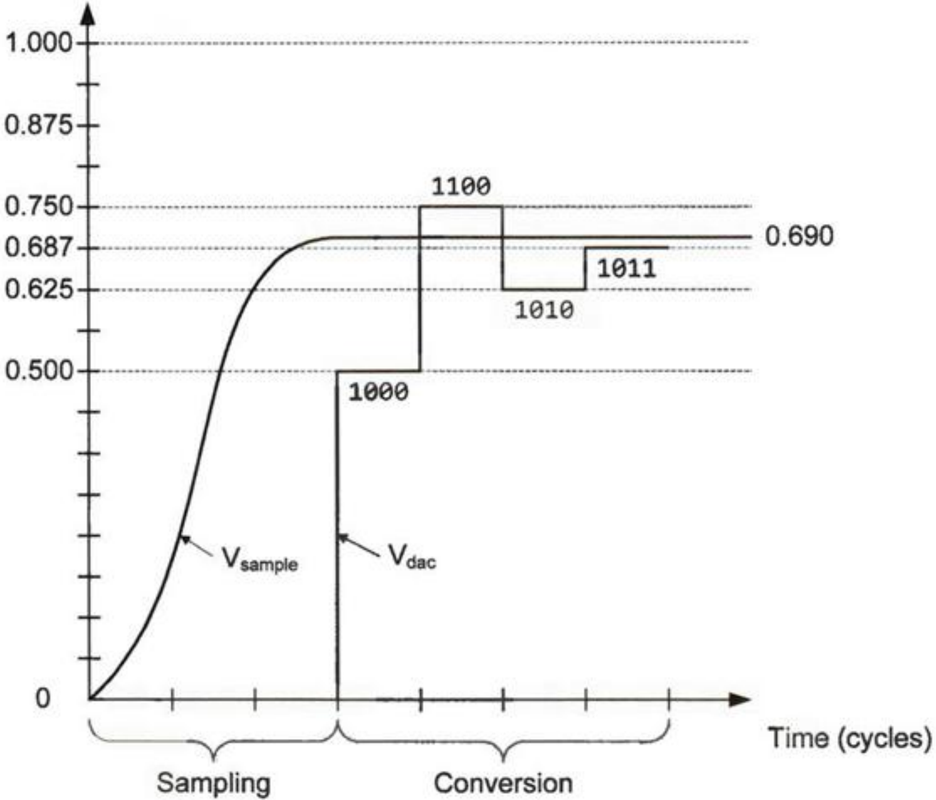


Fig: An example of four-bit successive-approximation (SAR) ADC. Suppose the input voltage is between 0 and 1 V.

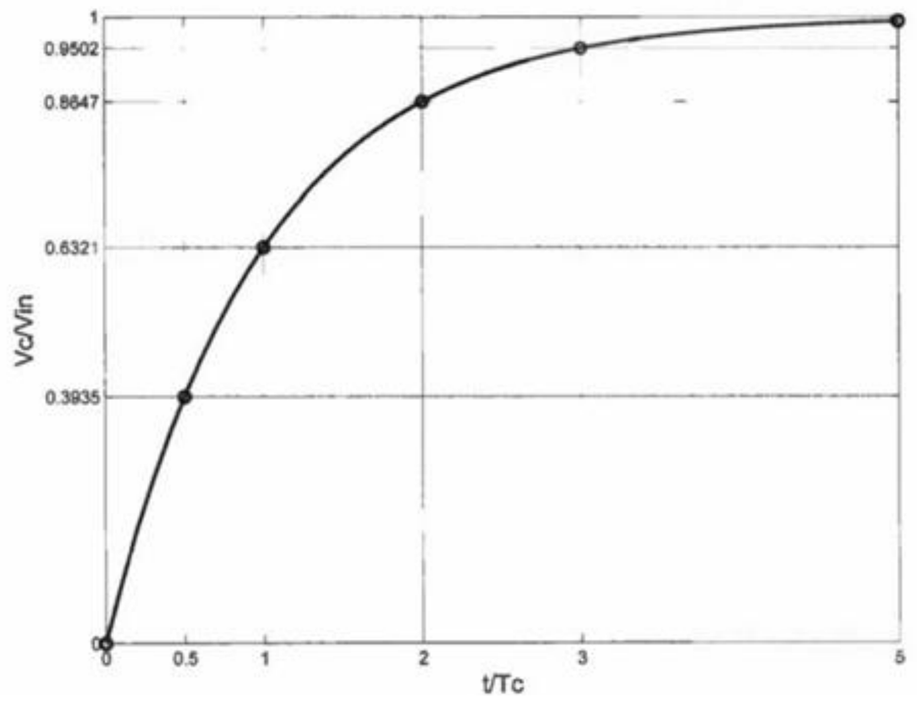
Sampling and Hold

The **sample-and-hold amplifier (SHA)** includes a switched capacitor and one operational amplifier, which is used to sample the analog input voltage V_{in} and hold the value over a certain amount of time for subsequent ADC processing. The sampling and hold circuit is a simple resistor-capacitor circuit. When the switch is closed, the voltage across the capacitor V_c increases exponentially.

$$V_C(t) = V_{in} \times \left(1 - e^{-\frac{t}{T_C}}\right) \qquad \text{where } T_c = (R_{in} + R_{adc}) \times C_{adc}$$

The input voltage V_{in} cannot be sampled instantly. For example, when the switch is closed for a time period of $3T_c$, the voltage across the capacitor V_c is only 95.02% of the input voltage V_{in}.

Sampling and Hold



To achieve accurate analog-to-digital conversion, software should let the capture switch close for enough time. The amount of time that the capture switch remains closed is called sampling time.

Fig: The change of the ratio of Vc (voltage across the capacitor) to Vin (Input voltage)

ADC Conversion Time

If the ADC has a resolution of n bits, the conversion time of successive-approximation takes n cycles to complete.

$$T_{conversion} = Sampling\ Time + Channel\ Conversion\ Time$$

For 12-bit ADC conversion, if the sampling time is set to 4 cycles and the ADC clock is set to 16 MHz, then we have,

$$T_{conversion} = 4 + 12 = 16\ cycles\ = 1\mu s$$